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COURSE CODE: SBT-DF202

LAB NAME: LAB4 STEGANOGRAPHY AND HIDDEN DATA DETECTION

DATE: 05/09/26

INSTRUCTOR: AMINU IDRIS

SCENARIO

Your practical context

You are acting as a junior digital forensic trainee. During a basic evidence review, an image file is suspected to contain hidden information. Your task is to understand how steganography works, prepare and preserve a carrier image, hide a controlled evidence note inside the image, extract the concealed file using the correct password, compare file hashes, and test a password-recovery workflow using a controlled wordlist.

All activities must be performed only on the authorised lab image and files created by you within the training environment.

OBJECTIVES

What you must demonstrate

By completing this lab, students will demonstrate their ability to explain the difference between steganography and encryption, describe insertion and substitution-based hiding techniques including least significant bits (LSB), prepare a carrier image and secret evidence file, use Steghide to embed and extract concealed data, calculate and compare cryptographic hashes, perform a controlled password dictionary test using StegSeek, and document all activities in a repeatable forensic report

INSTRUCTOR BRIEFING

Laboratory guidance

Before beginning this lab, review the **Module 8 lessons and study material on Steganography and Hidden Data Analysis**. You must understand the concepts of hidden information, substitution, least significant bits, bitmap images, Steghide, extraction and password recovery before performing the practical tasks.

Required Carrier Image

Download the authorised lab BMP image:

Steganography Lab Carrier Image

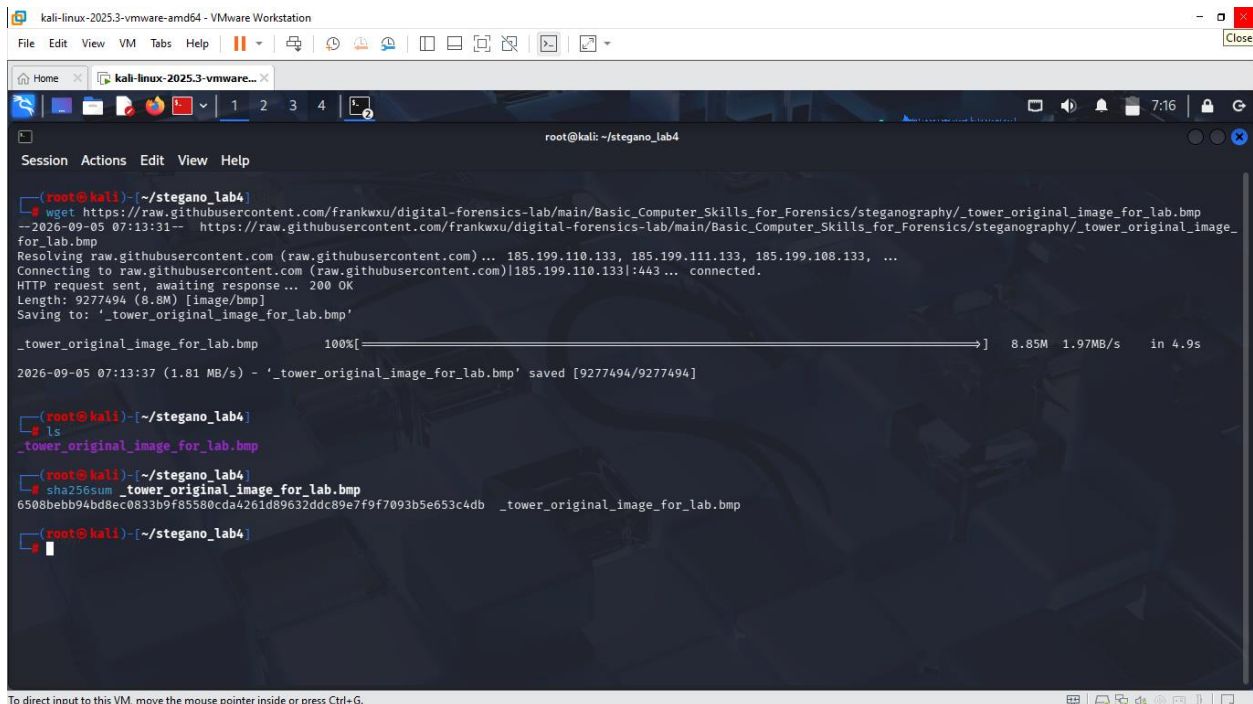
If browser download is required, use the link above. The source practical identifies this specific image as the lab carrier.

Part A — Prepare and Preserve the Evidence

Create a dedicated working folder, download the carrier image, confirm its file type and calculate its SHA-256 hash.

Preserve the original downloaded image. Do not overwrite it during the embedding exercise.

ANSWER:



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/stegano_lab4
Session Actions Edit View Help
root@kali) ~/stegano_lab4
$ wget https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/steganography/_tower_original_image_for_lab.bmp
--2026-09-05 07:13:31-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Basic_Computer_Skills_for_Forensics/steganography/_tower_original_image_for_lab.bmp
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.110.133, 185.199.111.133, 185.199.108.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.110.133|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 9277494 (8.8M) [image/bmp]
Saving to: '_tower_original_image_for_lab.bmp'

_tower_original_image_for_lab.bmp 100%[=====] 8.85M 1.97MB/s in 4.9s
2026-09-05 07:13:37 (1.81 MB/s) - '_tower_original_image_for_lab.bmp' saved [9277494/9277494]

root@kali) ~/stegano_lab4
$ ls
_tower_original_image_for_lab.bmp
root@kali) ~/stegano_lab4
$ sha256sum _tower_original_image_for_lab.bmp
6508bebb94bd8ec0833b9f85580cda4261d89632ddc89e7f9f7093b5e653c4db _tower_original_image_for_lab.bmp
root@kali) ~/stegano_lab4
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Part B — Create the Secret Evidence File

Create a text file named:

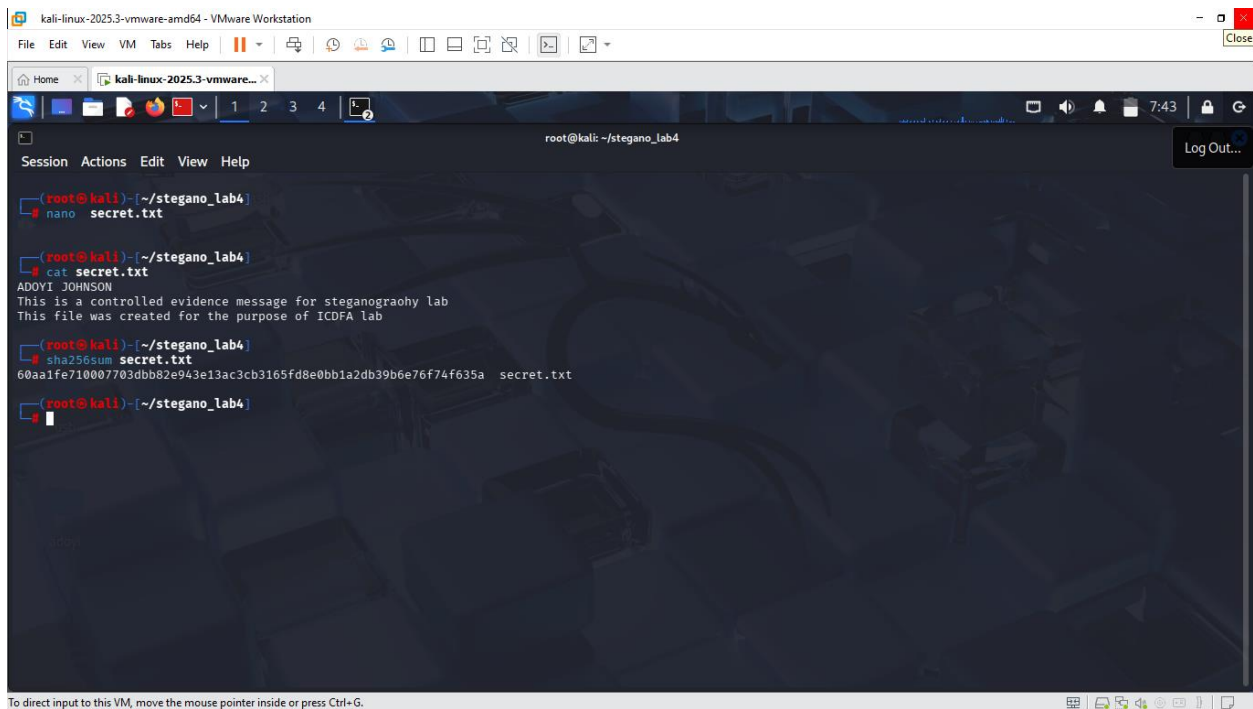
secret.txt

Include:

- Your full name
- A short controlled evidence message
- Confirmation that the file was created for the IC DFA lab

Calculate and record the SHA-256 hash of the file.

ANSWER:



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/stegano_lab4
Session Actions Edit View Help
(root@kali) ~/stegano_lab4
$ nano secret.txt
(root@kali) ~/stegano_lab4
$ cat secret.txt
AD0YI JOHNSON
This is a controlled evidence message for steganograohy lab
This file was created for the purpose of IC DFA lab
(root@kali) ~/stegano_lab4
$ sha256sum secret.txt
60aa1fe710007703dbb82e943e13ac3cb3165fd8e0bb1a2db39b6e76f74f635a secret.txt
(root@kali) ~/stegano_lab4
```

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Part C — Embed the Hidden File with Steghide

Use Steghide to embed secret.txt into a **new stego image**.

For the guided portion of the lab, use the lab password:

1234

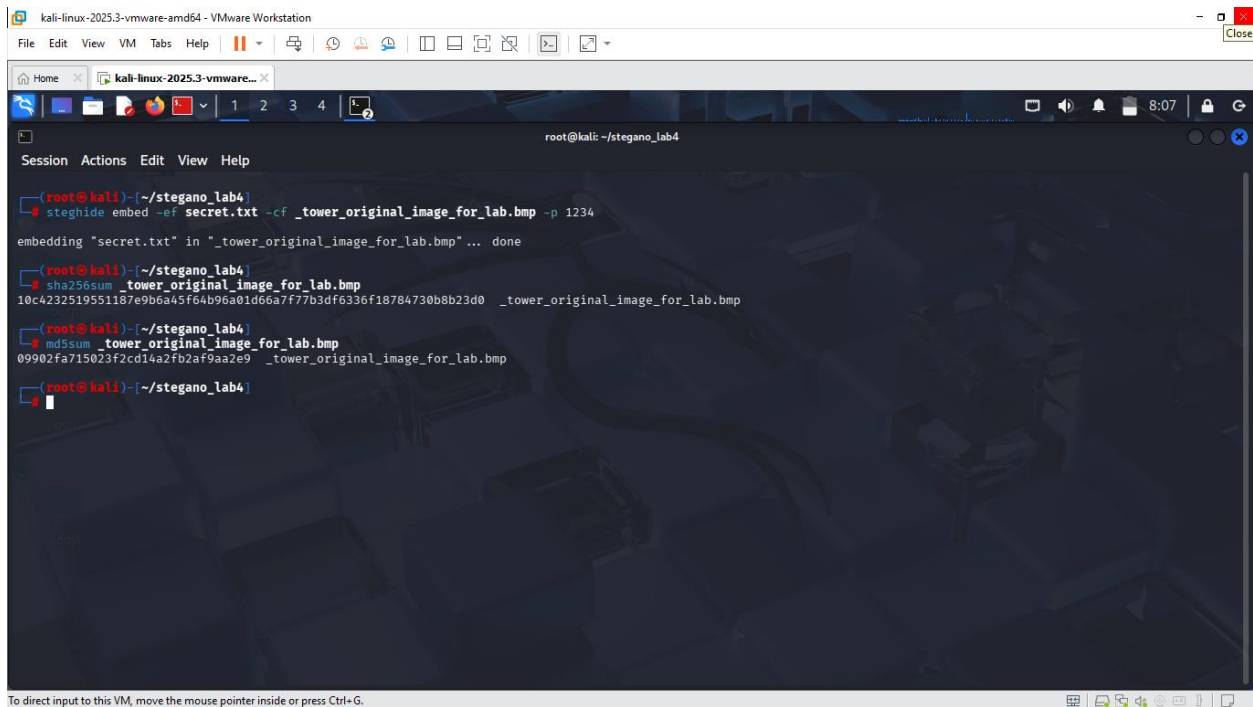
The source lab uses this password specifically so the later dictionary-recovery task can be completed.

Example structure:

```
steghide embed \-ef secret.txt \-cf tower_original_image_for_lab.bmp \-sf tower_stego_lab4.bmp \-p 1234
```

Calculate and record the hash of the resulting stego image.

ANSWER:



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-amd64
root@kali: ~/stegano_lab4
Session Actions Edit View Help
root@kali)~/(stegano_lab4)
# steghide embed -ef secret.txt -cf _tower_original_image_for_lab.bmp -p 1234
embedding "secret.txt" in "_tower_original_image_for_lab.bmp"... done
root@kali)~/(stegano_lab4)
# sha256sum _tower_original_image_for_lab.bmp
10c4232519551187e9b6a45f64b96a01d66a7f77b3df6336f18784730b8b23d0 _tower_original_image_for_lab.bmp
root@kali)~/(stegano_lab4)
# md5sum _tower_original_image_for_lab.bmp
09902fa715023f2cd14a2fb2af9aa2e9 _tower_original_image_for_lab.bmp
root@kali)~/(stegano_lab4)
#
```

Part D — Extract and Verify the Hidden Evidence

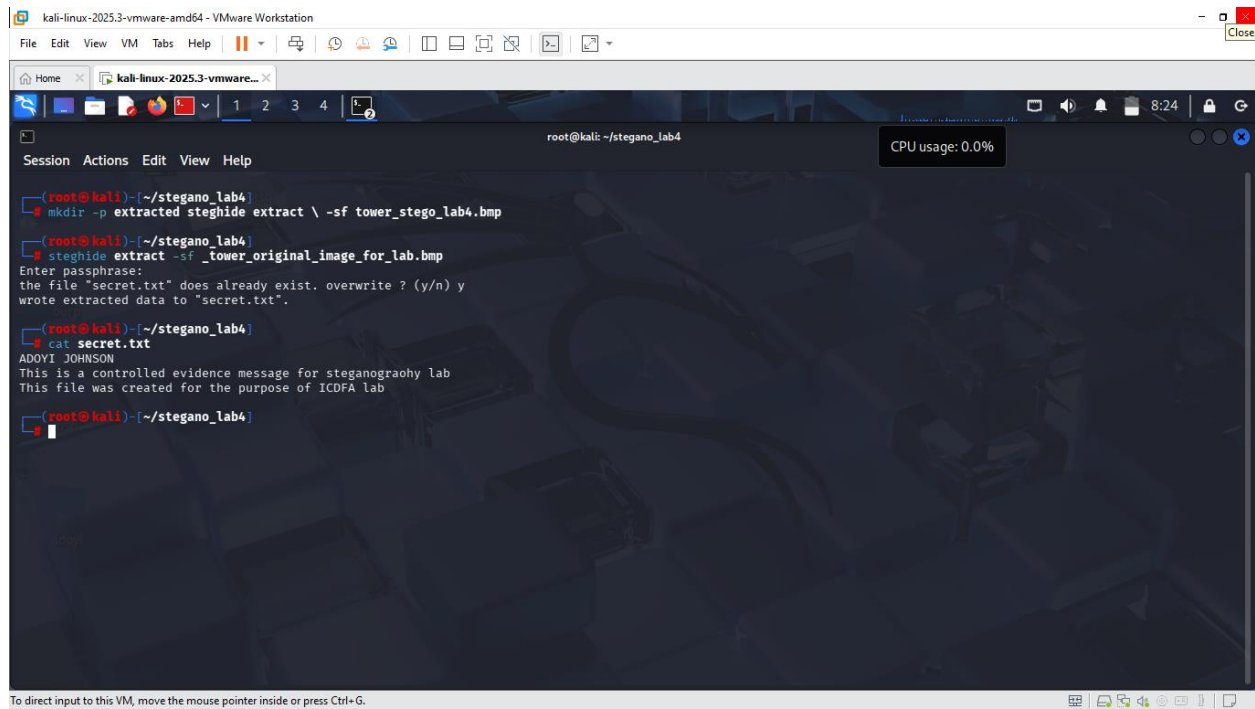
Extract the concealed file using the correct password.

Example:

```
mkdir -p extracted steghide extract \-sf tower_stego_lab4.bmp \-xf extracted/extracted_secret.txt \-p 1234
```

Compare the original and extracted files using diff and/or SHA-256 hashes. The recovered file should match the original if extraction is successful.

ANSWER:



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/stegano_lab4
CPU usage: 0.0%
Session Actions Edit View Help
root@kali) ~/stegano_lab4
# mkdir -p extracted steghide extract \ -sf tower_stego_lab4.bmp
root@kali) ~/stegano_lab4
# steghide extract -sf _tower_original_image_for_lab.bmp
Enter passphrase:
the file "secret.txt" does already exist. overwrite ? (y/n) y
wrote extracted data to "secret.txt".
root@kali) ~/stegano_lab4
# cat secret.txt
ADOYI JOHNSON
This is a controlled evidence message for steganograohy lab
This file was created for the purpose of ICofA lab
root@kali) ~/stegano_lab4
#
```

Part E — Compare Original and Stego Images

Compare:

- File sizes
- SHA-256 hashes
- Initial bytes using xxd
- Readable strings where applicable

ANSWER:

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-amd64
root@kali: ~/stegano_lab4
Session Actions Edit View Help
root@kali)~/stegano_lab4
$ ls -l _tower_original_image_for_lab.bmp tower_stego_lab4.bmp
-rw-rw-r-- 1 root root 9277494 Sep  5 07:59 _tower_original_image_for_lab.bmp
tower_stego_lab4.bmp:
total 0
root@kali)~/stegano_lab4
$ sha256sum _tower_original_image_for_lab.bmp tower_stego_lab4.bmp
10c4232519551187e9b6a45f64b96a01d66a7f77b3df6336f18784730b8b23d0 _tower_original_image_for_lab.bmp
sha256sum: tower_stego_lab4.bmp: Is a directory
root@kali)~/stegano_lab4
$ xxd _tower_original_image_for_lab.bmp tower_stego_lab4.bmp
xxd: tower_stego_lab4.bmp: Is a directory
root@kali)~/stegano_lab4
$
```

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-amd64
root@kali: ~/stegano_lab4
Session Actions Edit View Help
root@kali)~/stegano_lab4
$ xxd _tower_original_image_for_lab.bmp tower_stego_lab4.bmp
xxd: tower_stego_lab4.bmp: Is a directory
root@kali)~/stegano_lab4
$ xxd _tower_original_image_for_lab.bmp
00000000: 424d 3699 0400 0000 0000 3600 0000 2800  BM6 .....6 ... (
00000010: 0000 5545 0000 d904 0000 0400 0000 0000  ..U .. . . . .
00000020: 0000 0000 0000 1200 0000 1200 0000 0000  .... . . . .
00000030: 0000 0000 0000 4c22 3528 2838 4c20 381f  .... "5 #8 8
00000040: 283a 2425 3f28 2945 2729 482a 2b4d 2a2c  :$*(JE)H+H+
00000050: 4a29 2a4f 282e 5127 3052 2831 5328 325a  N).O(.Q'OR(1S(2T
00000060: 2f39 5b35 4163 3d4e 6f44 5979 465f 8147  /9[5Ac-NoDYyF _G
00000070: 6687 4669 8045 6809 4468 8045 6785 4a69  f Fl Eh Dh Eg JI
00000080: 8042 5f7e 3b57 7637 4e6e 3243 6432 3a60  B _;Wv7Mn2Cd2>
00000090: 313a 5c31 375a 3739 5c37 395c 3b3b 5f3b  1:\17Z79\79\;;
000000a0: 3b5f 3b3d 6037 395c 3637 5d35 365c 3536  ;_]='79\67]56\56
000000b0: 5c36 375d 3637 5d35 365c 3536 5c35 365c  \67]67]56\56\56\
000000c0: 3739 5c35 375a 3436 5935 375a 3436 5936  79\57Z46Y57Z46Y6
000000d0: 385b 3638 5b33 3558 3536 5832 3355 3233  8[68[35X56X23U23
000000e0: 5534 3557 3233 5531 3254 3334 5633 3456  U45W23U12T34V34V
000000f0: 3536 5834 3557 3334 562d 2e50 2e2f 512c  56X45W34V--P-/Q
00000100: 2d4f 2f30 522f 3052 3332 5231 3050 2f2e  -O/OR/OR32R10P/
00000110: 4a2e 2d4d 2f2e 4e2f 2a4e 2e2e 4c2d 2d4b  N.-M/.M/.N..L--K
00000120: 2f2f 4d2b 2b49 2e2e 4c2d 2d4b 2c2d 492c  //M+I..L--K,-I
00000130: 2d49 2a2b 472a 2b47 292c 4826 2945 2629  -I+G+G),H6)E6
00000140: 4527 2a46 262a 4327 2b44 292a 4427 2941  E'+F6+C'+D)*D')A
00000150: 282a 4226 2840 2729 4127 2941 2629 3e26  (*B6(a')A')A6>6
00000160: 293e 272a 3f24 273c 2b2c 412b 2c41 2a2b  >)*?5'<+,A+,A+
00000170: 4a28 293e 2728 3d27 283d 2829 3e28 293d  a(>)'(='(>(>)=
00000180: 2829 3d27 293b 2527 3923 2537 2325 3723  (=)';%9#%7#%7#
```

Explain why two images may look visually similar while their underlying digital content and hashes are different.

ANSWER:

In digital forensics, this is important because visual appearance alone cannot prove that two files are identical. Hash values are used to verify the exact digital integrity of a file.

Part F — Controlled Password Dictionary Test

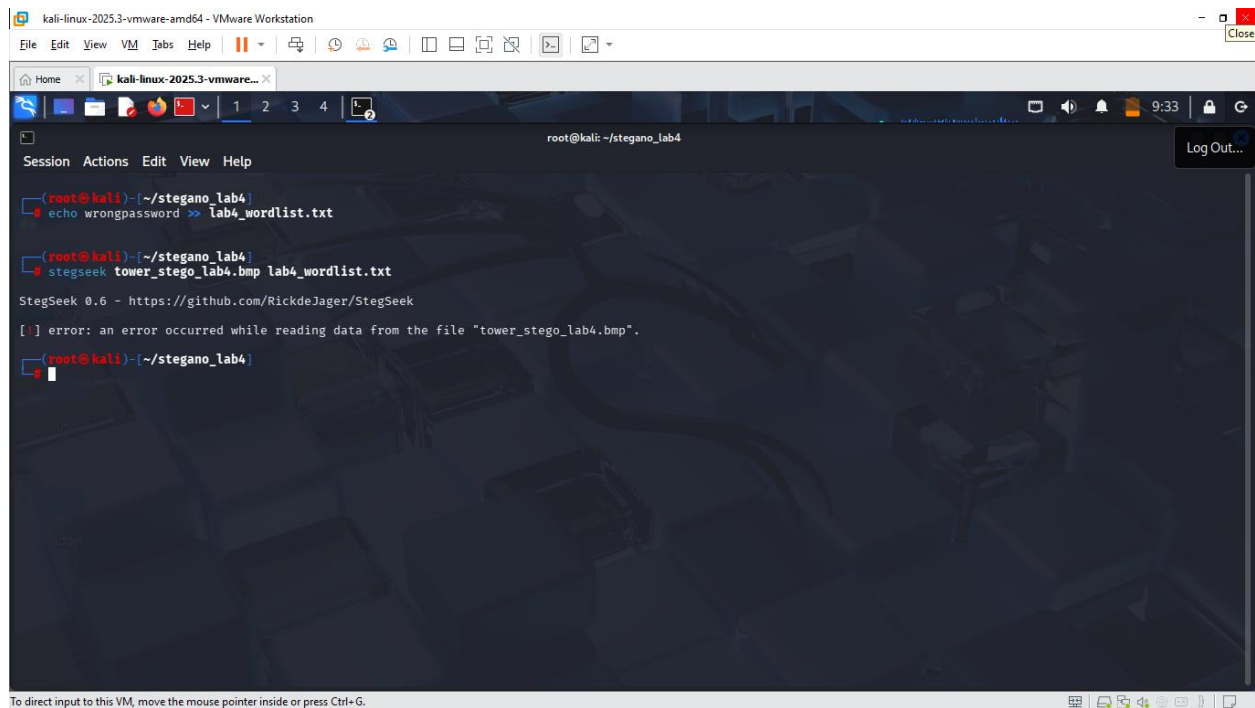
Create a small lab wordlist that contains the known password 1234.

Use StegSeek against the stego image:

```
stegseek tower_stego_lab4.bmp lab4_wordlist.txt
```

Document the result.

If StegSeek is unavailable or fails in your environment, document the error clearly and complete the Steghide embedding/extraction workflow successfully instead.

A screenshot of a Kali Linux virtual machine running in VMware Workstation. The terminal window shows the user at the root prompt in the directory ~/stegano_lab4. The user has created a wordlist file named lab4_wordlist.txt containing the password 'wrongpassword'. They then run the command 'stegseek tower_stego_lab4.bmp lab4_wordlist.txt'. The output shows the StegSeek version (0.6) and a GitHub link, followed by an error message: '[] error: an error occurred while reading data from the file "tower_stego_lab4.bmp".' The terminal window has a dark background with a cityscape pattern. The VMware interface shows the VM is named 'kali-linux-2025.3-vmware-amd64' and the session is titled 'root@kali: ~/stegano_lab4'.

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
Session Actions Edit View Help
root@kali: ~/stegano_lab4
Log Out...

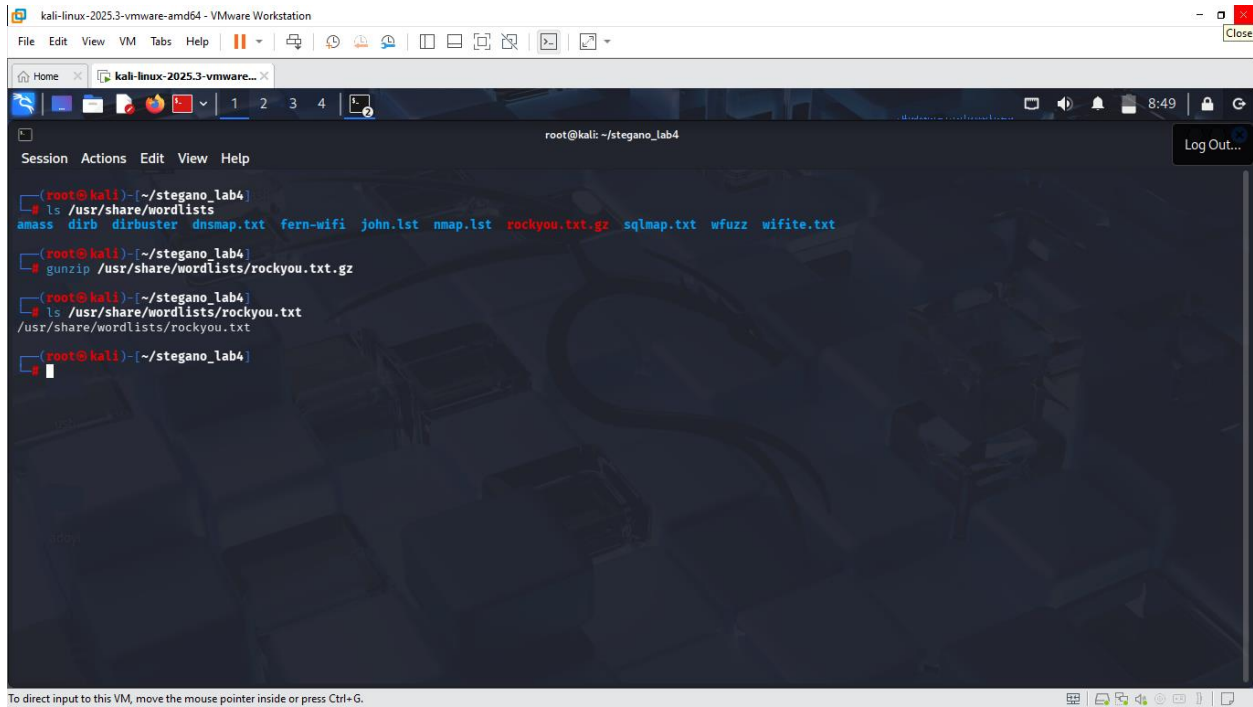
(root@kali)~/stegano_lab4
# echo wrongpassword >> lab4_wordlist.txt

(root@kali)~/stegano_lab4
# stegseek tower_stego_lab4.bmp lab4_wordlist.txt

StegSeek 0.6 - https://github.com/RickdeJager/StegSeek

[ ] error: an error occurred while reading data from the file "tower_stego_lab4.bmp".

(root@kali)~/stegano_lab4
#
```



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-amd64
root@kali: ~/stegano_lab4
Session Actions Edit View Help
Log Out...

(root@kali)~/stegano_lab4
# ls /usr/share/wordlists
amass dirb dirbuster dnsmap.txt fern-wifi john.lst nmap.lst rockyou.txt.gz sqlmap.txt wfuzz wifite.txt

(root@kali)~/stegano_lab4
# gunzip /usr/share/wordlists/rockyou.txt.gz

(root@kali)~/stegano_lab4
# ls /usr/share/wordlists/rockyou.txt
/usr/share/wordlists/rockyou.txt

(root@kali)~/stegano_lab4
#
```

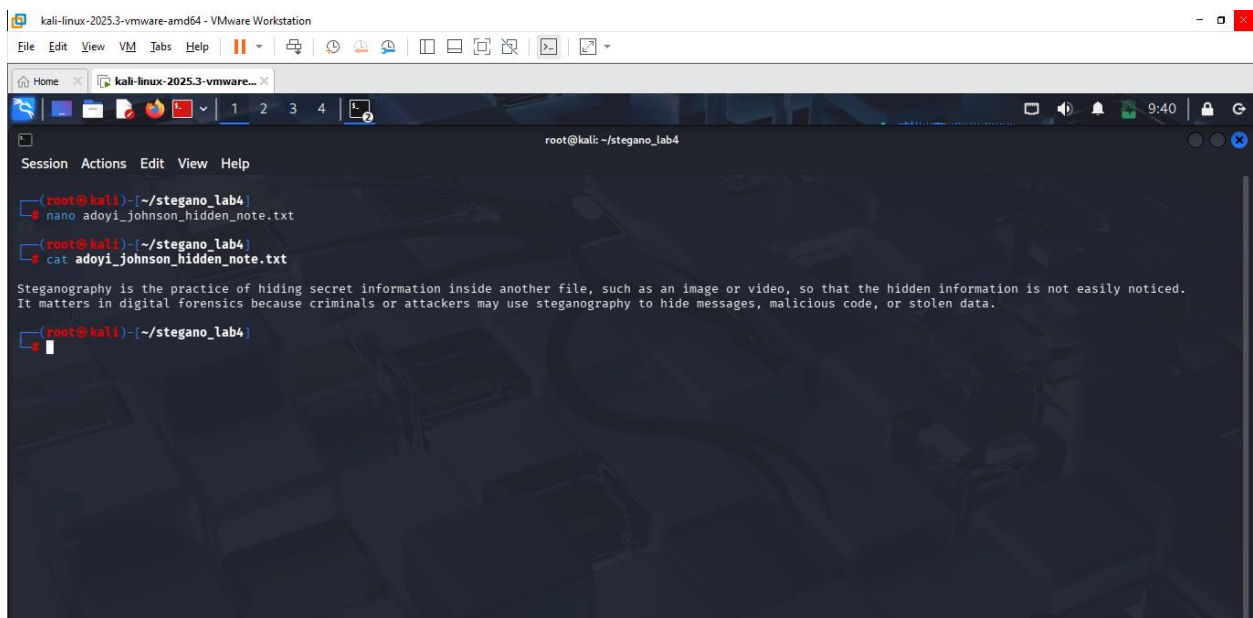
To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

Part G — Independent Student Task

Create your own hidden evidence file:

firstname_lastname_hidden_note.txt

Write a short explanation of what steganography is and why it matters in digital forensics.



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware-amd64
root@kali: ~/stegano_lab4
Session Actions Edit View Help
Log Out...

(root@kali)~/stegano_lab4
# nano adoyi_johnson_hidden_note.txt

(root@kali)~/stegano_lab4
# cat adoyi_johnson_hidden_note.txt

Steganography is the practice of hiding secret information inside another file, such as an image or video, so that the hidden information is not easily noticed.
It matters in digital forensics because criminals or attackers may use steganography to hide messages, malicious code, or stolen data.

(root@kali)~/stegano_lab4
#
```


Steganography is the practice of hiding secret information inside another file, such as an image or video, so that the hidden information is not easily noticed.

It matters in digital forensics because criminals or attackers may use steganography to hide messages, malicious code, or stolen data.

Then:

1. Embed it into the carrier image.
2. Use a password of your choice.
3. Extract it again.
4. Verify integrity using hashes.
5. Create a small wordlist containing your chosen password.
6. Test the recovery workflow.
7. Document every step with screenshots.

Required evidence

Submit one professional PDF forensic report together with the required supporting evidence files.

Required evidence should include:

Proof of Module 8 pre-lab review and environment preparation.

Original carrier image details and hash.

secret.txt and its hash.

Steghide embedding command and resulting stego image.

Successful extraction and hash/diff verification.

Original versus stego image comparison using hashes and xxd.

StegSeek/wordlist result or documented tool failure.

Student-created independent steganography evidence file and extraction proof.

Clear screenshots and command evidence throughout.

The source submission checklist requires the PDF report, secret file, stego image, extracted file, hash file, custom wordlist, and explanation of the LSB/substitution concept.

Lab access notes This practical forms part of the second laboratory assessment window. Students must review the Module 8 lessons and study material before beginning. Use only the authorised lab carrier image and files created personally for this exercise. Preserve the original carrier image, record cryptographic hashes, and document all embedding, extraction, comparison and password-testing activities.